159102 workshop – Week 10

1. Write a C++ program that reads positive real numbers into a vector (in a random order). Use a trailing value of -1. Sort the numbers into ascending order and display the sorted numbers. Your program must include the following two functions:

```
void sort(vector<float> & v);
void display(vector<float> & v);
```

2. In your sorting program above, change the first line of the `sort` function to be:

```
void sort(vector<float> v);
```

(*Note that you will also have to change the first line in the function definition*.)
Do not change any other code. Run the program. Explain what happens.

3. Discuss what you would need to do to change your sorting program above so that it uses an array and not a vector. What is the single biggest difference between the two programs? This is for discussion only – there is no need to actually write the program using an array.

4. The program below enters data for bank accounts and then searches to find a particular bank account. However, the program is not working because there are 6 errors in it. Identify the errors and state what needs to be done to correct each error.

```
#include <iostream>
#include <cstdio>  // for getchar
#include <vector>
using namespace std;

class bank_account {
private:
  int accnumber;
  string name;
  float balance;
public:
  void loaddata();
  void display();
};

void enteraccounts();

int main() {
  int search, total, num, i;
  enteraccounts();
  cout << "Which account do you want to display? ";
  cin >> search;
  total = accounts.size();
  for (i = 0; i < total; i++) {
    num = accounts[i].getnum();
    if (num != search) {
      accounts[i].display();
```

```
    }
  }
}

void enteraccounts() {
  // enter 3 accounts - this could be changed
  int i;
  bank_account temp;
  for (i = 0; i < 3; i++) {
    temp.loaddata();
  }
}

//--------------- methods for the bank_account ------------
void loaddata() {
  cout << "Enter the bank account number ";
  cin >> accnumber;
  cout << "Enter the name of the account ";
  getline(cin, name);
  cout << "Enter the balance, e.g. 32.45 ";
  cin >> balance;
}

void bank_account::display() {
  cout << "Data for Account " << accnumber << endl;
  cout << "Name: " << name << endl;
  cout << "Balance: $" << balance << endl << endl;
}
```

5. Assume that the program below has a function to display all items in the stack (similar to the "display and empty" function in the notes). Write down what values would be displayed and in what order.

```
stack<int> s;

int main() {
  s.push(27);
  s.push(83);
  s.pop();
  s.push(56);
  -- call function here to display all items
}
```